

Robust Angle-Uncertainty-Aware Pareto Waveform Design for Multi-User Multi-Target MIMO-ISAC

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I. INTRODUCTION

Integrated sensing and communication (ISAC) has been widely regarded as a promising technology for sixth-generation (6G) wireless networks, since it enables wireless communication and radar sensing to share the same hardware platform, waveform, spectrum, and power resources [1]. Compared with separately designed communication and sensing systems, ISAC can improve spectral efficiency, reduce hardware cost, and provide both data transmission and environmental perception capabilities. These features are particularly important for future applications such as autonomous driving, smart transportation, unmanned aerial vehicles, smart cities, and location-aware wireless services [1].

Among different ISAC research directions, waveform design plays a fundamental role because the transmitted signal must simultaneously support reliable communication and accurate sensing. However, communication and sensing usually have different waveform requirements. Communication systems prefer waveforms that support high data rate and low multi-user interference, while sensing systems require good beam-pattern, high angular resolution, and robust target illumination. Therefore, it is difficult to optimize both functions simultaneously using a single fixed waveform [1].

Recent studies have shown that joint waveform optimization is a flexible and powerful approach for ISAC systems by incorporating communication and sensing metrics into a unified optimization problem [1]. For multi-user multi-target MIMO-ISAC systems, Pareto-optimal waveform design has been proposed to characterize the trade-off between communication sum rate and sensing beampattern matching performance [2], [3]. By using constructive interference (CI), multi-user interference can be converted into useful signal power, and by applying a weighted Chebyshev or weighted Tchebycheff transformation, the communication-sensing trade-off can be formulated as a Pareto optimization problem [2], [3].

However, most existing Pareto waveform design methods assume that the sensing target angles are perfectly known when constructing the desired beampattern [2], [3]. This assumption may not hold in practical ISAC sensing or tracking scenarios, where target direction-of-arrival (DoA) estimates are affected by noisy observations, target mobility, limited angular resolution, hardware imperfections, and tracking errors [4]. As a result, a waveform optimized only for nominal target directions may suffer sensing performance degradation when the actual target angle deviates from the assumed angle.

Motivated by this issue, this project proposes a Robust Angle-Uncertainty-Aware Pareto Waveform Design for Multi-User Multi-Target MIMO-ISAC. The main idea is to extend

the existing Pareto waveform design framework by replacing the nominal sensing beampattern matching metric with a robust sensing metric evaluated over bounded target angle uncertainty regions. The proposed design aims to obtain Pareto-optimal waveforms that balance communication sum rate and worst-case sensing robustness under target angle mismatch.

II. PROBLEM AND PROPOSED APPROACH

This project studies a downlink multi-user multi-target MIMO-ISAC system, where a multi-antenna Dual-Function Radar-Communication (DFRC) base station simultaneously serves multiple communication users and senses multiple point-like targets. Existing Pareto waveform design methods usually assume that target angles are perfectly known, which may cause sensing performance degradation when practical DoA estimation errors exist. To address this issue, this project considers bounded target angle uncertainty and proposes a robust angle-uncertainty-aware Pareto waveform design. The communication objective is to maximize the downlink sum rate under constructive interference constraints, while the sensing objective is to minimize the worst-case beampattern matching error over all possible target angles within the uncertainty intervals. The robust multi-objective problem is transformed into a scalar problem using the weighted Tchebycheff method, and SDR/SCA techniques are applied to handle the non-convex constraints. By sweeping different communication-sensing weights, the robust Pareto front can be obtained and compared with the conventional non-robust design.

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